AML TRANSACTION & FRAUD ANALYSIS DASHBOARD

Using Microsoft Power BI – Project Report

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Tool Used: Microsoft Power BI Desktop

Introduction

For this project, I developed an AML (Anti-Money Laundering) Transaction & Fraud Analysis Dashboard using Microsoft Power BI. I selected this topic because transaction monitoring and fraud detection are important areas in the banking and financial services industry. Financial institutions need to monitor transactions, identify suspicious activities, and analyse patterns that may indicate potential financial crime.

The main purpose of this dashboard was to transform transaction-related data into an interactive visual report that makes it easier to understand transaction volumes, fraud distribution, transaction values, and alert patterns.

The dashboard includes KPI cards and different visualisations to examine transaction activity and fraud-related information. It also provides a way to explore transaction trends over time and compare transaction values according to fraud status.

Visual charts are important in financial analysis because they make large amounts of data easier to understand. Instead of examining individual transaction records manually, users can observe patterns and comparisons through charts, tables, and summary figures.

In this project, I used Microsoft Power BI to create an interactive dashboard with connected visualisations and filters. This project helped me understand how business intelligence tools can support transaction monitoring and financial crime analysis.

The dashboard is designed to support the analysis of transaction data. However, identifying a fraudulent transaction through a dashboard does not, by itself, establish that money laundering or fraud has occurred. Further investigation is required to verify suspicious activities.

Objectives

The main objectives of this project were:

1. To analyse transaction activity using Microsoft Power BI.
2. To identify key performance indicators (KPIs) that summarise transaction activity and fraud-related information.
3. To examine transaction volumes across different timestamps and identify changes over time.
4. To compare the distribution of transactions according to fraud status.
5. To analyse fraudulent transaction volumes over time.

6. To examine transaction values according to fraud status.
7. To compare alerts across different alert types.
8. To create an interactive dashboard that allows users to filter and explore the available data.
9. To present transaction-related information through interconnected visualisations.
10. To understand how data visualisation can support transaction monitoring and preliminary fraud-risk analysis.

Dataset Description

The dataset used in this project relates to AML transaction and fraud analysis. The Power BI report contains visualisations related to transaction activity, timestamps, fraud status, transaction values, and alert types.

The following data elements are reflected in the dashboard:

- **Timestamp:** Represents the time associated with a transaction and supports the analysis of transaction activity over time.
- **Fraud Status:** Identifies the fraud-status category associated with a transaction, allowing comparisons between the available categories.
- **Transaction Value:** Represents the monetary value associated with a transaction and helps compare transaction values across fraud-status categories.
- **Alert Type:** Represents the category of an alert and helps analyse the distribution of alerts across different types.

These fields provide a foundation for analysing transaction patterns and fraud-related indicators.

The exact source schema, complete list of columns, total number of records, and data types should be verified in Power BI's Data and Model views before the final report is submitted.

Data Preparation

Before analysing the data through visualisations, it is important to ensure that the dataset is suitable for analysis. Data preparation helps improve the reliability of the results and reduces errors in calculations.

The following checks and preparation steps are relevant to this project:

1. **Importing the dataset:** The transaction dataset is loaded into Microsoft Power BI Desktop.
2. **Checking column names:** The available columns are examined to understand the information contained in the dataset.
3. **Checking data types:** Numeric fields, categorical fields, and timestamp fields are checked to ensure they have appropriate data types.
4. **Checking timestamps:** The timestamp column is checked to ensure that Power BI recognises it as a date/time field. This is important for analysing transaction trends over time.

5. **Checking transaction values:** The transaction-value field is checked to ensure that it is stored as a numeric value so that calculations and comparisons can be performed.
6. **Checking fraud status:** The fraud-status categories are examined to ensure that the values are consistent and suitable for grouping transactions.
7. **Checking alert types:** The alert-type field is examined to identify the categories available for analysis.
8. **Checking missing values:** The dataset is checked for blank values in important columns. Missing values should be handled according to their meaning rather than automatically replacing them.
9. **Creating DAX measures:** Measures are created to calculate transaction counts, transaction values, and fraud-related indicators for the KPI cards and charts.

These preparation steps help establish a consistent foundation for the dashboard. The specific cleaning operations performed should be confirmed against the actual Power Query steps in the PBIX file.

DAX Measures

DAX (Data Analysis Expressions) is the formula language used in Power BI to perform calculations and create measures.

In an AML dashboard, DAX measures can summarise transaction activity, calculate transaction values, and provide fraud-related indicators.

The following are suggested measures for this project. They are illustrative formulas, so the table and column names must be replaced with the exact names in the Dataset 5 data model.

Assumed table name: TransactionData

5.1 Total Transaction Count

Total Transaction Count =

```
COUNTROWS('TransactionData')
```

This measure counts the transaction records in the current filter context. It can be used in a KPI card to summarise the number of records included in the analysis.

5.2 Total Transaction Value

Total Transaction Value =

```
SUM('TransactionData'[TransactionValue])
```

This measure calculates the total transaction value. It can be used to compare the monetary value associated with different fraud-status categories.

5.3 Fraudulent Transaction Count

Fraudulent Transaction Count =

```
CALCULATE(
    COUNTROWS('TransactionData'),
```

'TransactionData'[FraudStatus] = "Fraud"

)

This measure counts the records classified as "Fraud", assuming that this exact category exists in the dataset.

If the actual fraud-status column uses different labels, the filter condition must be modified accordingly.

5.4 Fraudulent Transaction Value

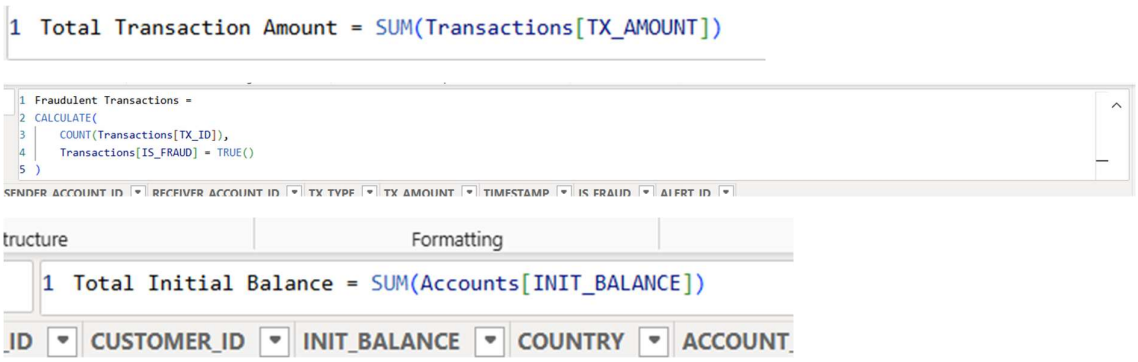
Fraudulent Transaction Value =

CALCULATE(

SUM('TransactionData'[TransactionValue]),

'TransactionData'[FraudStatus] = "Fraud"

)



This measure calculates the total value of transactions classified as fraudulent. It can help compare the monetary value of transactions across different categories.

Important: These formulas are proposed measures for the report, not verified DAX expressions extracted from the existing PBIX file. The actual table name, column names, and fraud-status labels must be checked before using them.

Data Components and Visuals Used

After examining the dashboard structure, I identified several visual components used to analyse transaction activity and fraud-related information. Each visual provides a different perspective on the available data.

The dashboard contains KPI cards, line charts, a donut chart, a clustered bar chart, a clustered column chart, and slicers.

6.1 KPI Cards

KPI cards are used to present important numerical values in a clear and concise format.

The dashboard includes multiple KPI cards that provide a quick overview of the data.

Depending on the measures assigned to the cards, suitable indicators for this dashboard include:

- **Total Transaction Count:** Displays the number of transaction records.
- **Total Transaction Value:** Displays the total monetary value of transactions.
- **Fraudulent Transaction Count:** Displays the number of transactions classified as fraudulent.
- **Fraudulent Transaction Value:** Displays the total value of transactions classified as fraudulent.

These are suitable KPI definitions for an AML dashboard. The exact measures currently assigned to the existing cards should be verified in Power BI.

KPI cards are useful because they allow users to understand the overall transaction activity without examining every chart individually.



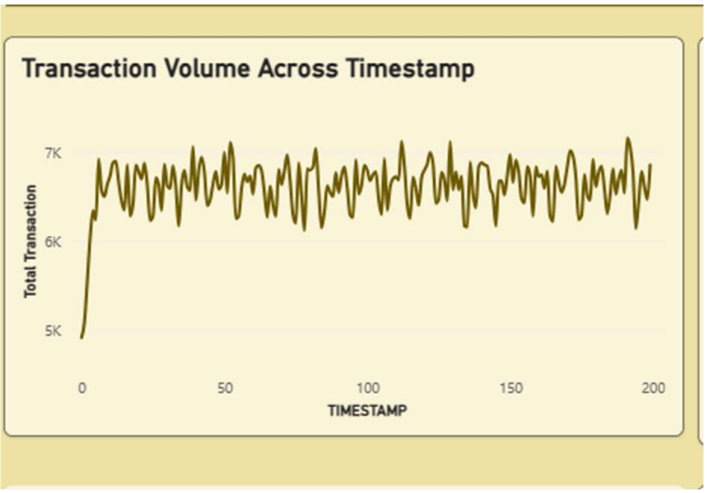
6.2 Transaction Volume Across Timestamp

I used a line chart titled **"Transaction Volume Across Timestamp"** to examine transaction activity over time.

The timestamp is used as the time-based field, while transaction volume represents the number of transaction records.

This visual helps users observe changes in transaction activity across the selected period. Peaks and sudden changes in transaction volume may indicate periods that require further examination.

However, a sudden increase in transaction volume does not automatically indicate fraudulent activity. The change must be interpreted alongside other information.



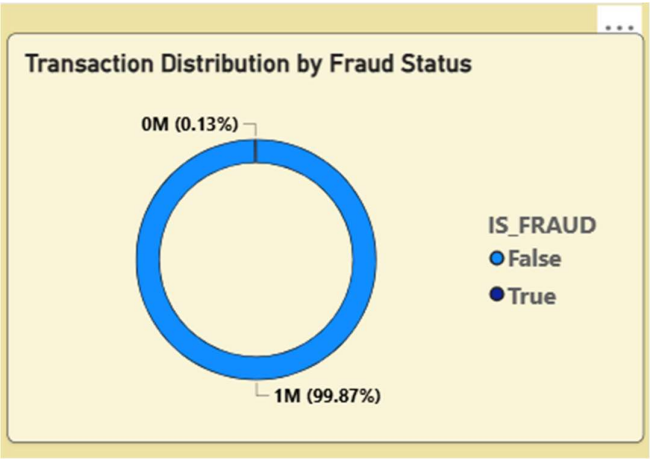
6.3 Transaction Distribution by Fraud Status

I used a donut chart titled **"Transaction Distribution by Fraud Status"** to compare the distribution of transactions across the available fraud-status categories.

This visual helps users understand how transaction records are distributed between the categories represented in the dataset.

The chart can provide a quick overview of the proportion of records classified under each status.

It is important to interpret this visual carefully. The distribution reflects the labels present in the dataset and does not independently establish the actual prevalence of fraud in the wider financial system.



6.4 Fraudulent Transaction Volume Across Timestamp

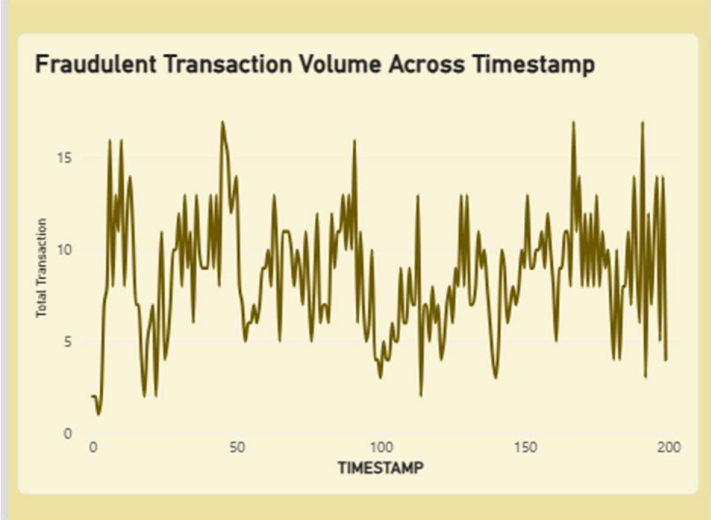
I used a line chart titled **"Fraudulent Transaction Volume Across Timestamp"** to examine changes in the volume of transactions classified as fraudulent over time.

The timestamp provides the time axis, while the fraudulent transaction count provides the measure.

This visual can help users identify periods when the number of transactions classified as fraudulent increases or decreases.

For example, an increase in the chart may indicate a period that deserves closer investigation. However, the reason for the increase must be established through further analysis of the underlying transactions.

This chart is useful for understanding the time-based distribution of fraud-labelled records.



6.5 Alerts by Alert Type

I used a clustered bar chart titled **"Alerts by Alert Type"** to compare the number of alerts across different alert categories.

The alert type is used as the category, and the relevant alert count is used as the value.

This visual helps users identify which alert categories account for a larger or smaller number of records in the dataset.

It can also support a comparison of alert patterns and help identify categories that may require further review.

A higher alert count does not necessarily mean that a category contains more confirmed fraud. Alerts may require investigation before a conclusion can be reached.

6.6 Total Transaction Value by Fraud Status

I used a clustered column chart titled **"Total Transaction Value by Fraud Status"** to compare transaction values across fraud-status categories.

The fraud-status category is used on the axis, and the total transaction value is used as the numerical measure.

This visual helps users compare the monetary value associated with the different categories represented in the dataset.

It provides a financial perspective in addition to the transaction-count analysis.

For example, two categories may contain different numbers of transactions but have similar total transaction values. Examining both transaction counts and monetary values can therefore provide a more complete understanding of the data.

The actual comparison depends on the values shown in the dashboard.

6.7 Interactive Slicers

The dashboard includes slicers that allow users to filter the available information.

One of the visible slicers is titled **"Fraud status"**. The report also contains another slicer whose exact field should be verified in Power BI.

Slicers are useful because they allow users to examine a selected category without changing the underlying dataset.

For example, selecting a fraud-status category can help users examine the associated transaction trends and transaction values, provided the visuals are connected through the report's filter interactions.

The available slicers and their effects should be tested in Power BI to confirm that the relevant charts and KPI cards respond correctly.

6.8 Dashboard Interactivity

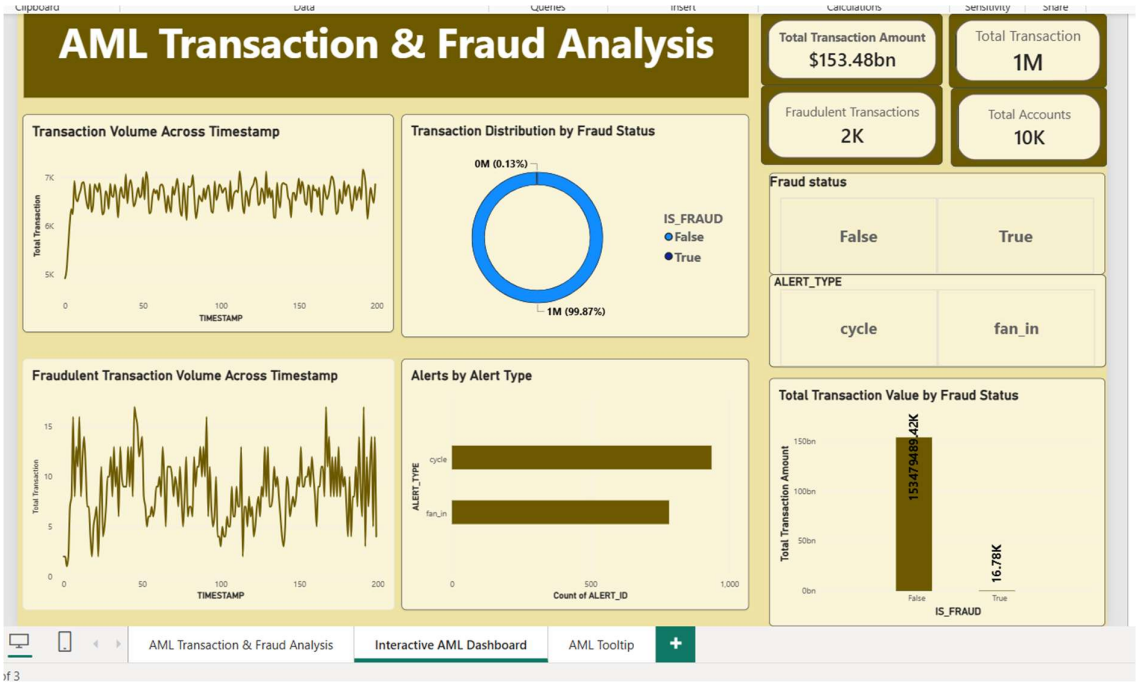
The dashboard combines KPI cards, line charts, a donut chart, a clustered bar chart, a clustered column chart, and slicers.

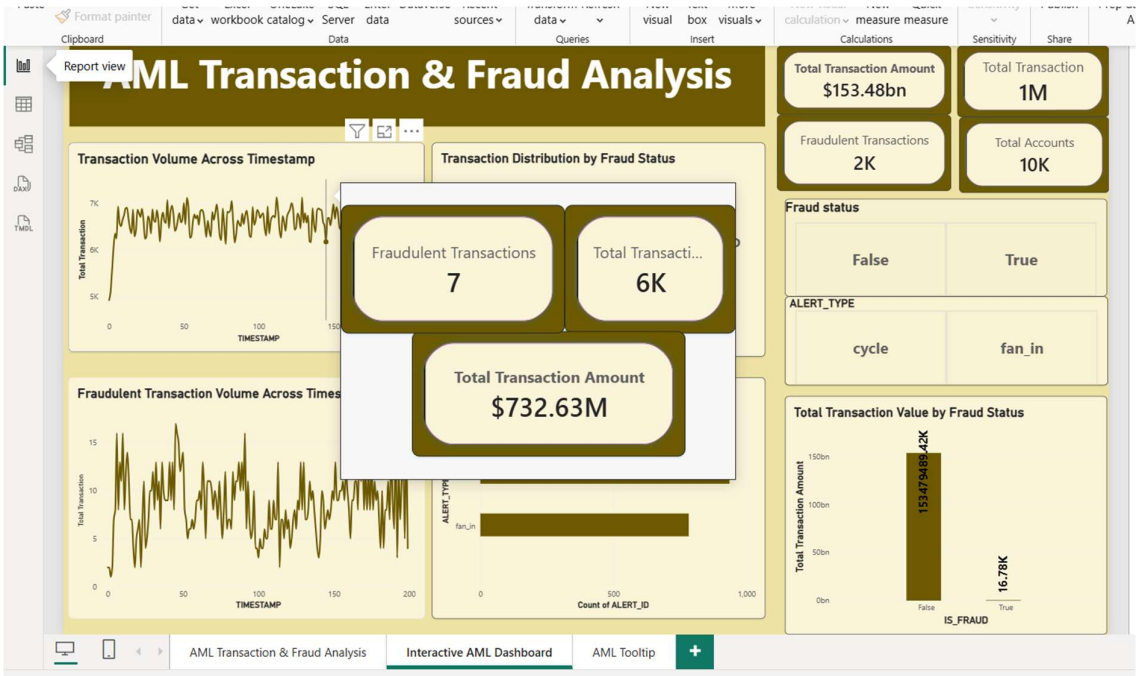
These visual components provide different perspectives on transaction activity:

- KPI cards summarise important numerical indicators.
- Line charts show changes in transaction activity over time.
- The donut chart compares the distribution of transactions by fraud status.
- The clustered bar chart compares alert categories.
- The clustered column chart compares transaction values by fraud status.
- Slicers allow users to filter the displayed information.

Together, these components help users move from an overall summary to a more detailed examination of transaction patterns.

The dashboard's actual filter interactions should be tested to confirm that selections in one visual update the intended related visuals.





Key Findings

The dashboard provides a visual framework for examining transaction activity and fraud-related patterns. The following observations describe what the available visualisations can help analyse.

1. Transaction Activity Over Time

The transaction-volume line chart allows users to examine changes in transaction activity across timestamps. Periods with unusually high or low activity can be identified for further investigation.

2. Distribution by Fraud Status

The donut chart allows users to compare the distribution of transactions across the fraud-status categories represented in the dataset.

This helps provide an overview of the classification of transaction records.

3. Changes in Fraud-Labelled Transaction Volume

The fraudulent transaction-volume chart allows users to examine how the number of records classified as fraudulent changes over time.

Any noticeable increase or decrease should be investigated further to understand the underlying reason.

4. Comparison of Alert Categories

The alert-type bar chart helps users compare the number of alerts across categories.

This can help identify which categories contribute more heavily to the alerts represented in the dataset.

5. Transaction Value Comparison

The clustered column chart allows users to compare the total transaction value associated with each fraud-status category.

This provides a monetary perspective that complements transaction-count analysis.

6. Importance of Combining Different Visuals

The dashboard presents transaction counts, transaction values, fraud-status distribution, and alert categories through different visualisations.

Examining these indicators together provides a broader view than relying on a single chart.

7. Importance of Interactive Filtering

The slicers allow users to explore selected categories and examine the corresponding information in the report. This makes the dashboard more flexible for analysis.

8. Need for Further Investigation

The dashboard can highlight patterns that deserve attention, but it cannot independently establish that a transaction is illegal or that money laundering has occurred.

Additional investigation, supporting information, and appropriate verification are required before reaching such conclusions.

Note: These are analytical observations supported by the dashboard's structure, not verified numerical findings. Exact figures, such as the total number of transactions, the number of fraud-labelled records, the total transaction value, and the most frequent alert type, should be calculated from the actual dataset before being included in the final submitted report.

Conclusion

Through this project, I learned how Microsoft Power BI can be used to convert transaction-related data into an interactive dashboard for AML and fraud analysis.

I explored how KPI cards, line charts, a donut chart, bar charts, column charts, and slicers can be used to present different aspects of transaction activity.

The dashboard provides a framework for analysing transaction volumes over time, comparing transactions according to fraud status, examining transaction values, and understanding the distribution of alerts across categories.

This project also helped me understand the importance of data preparation, suitable visualisation selection, and interactive filtering when working with financial data.

From a FinTech perspective, transaction monitoring and fraud analysis are important applications of data analytics in financial services. Power BI can help present transaction patterns in a clear and accessible format, making it easier for analysts to identify areas that may require further examination.

Overall, this project gave me practical exposure to the use of business intelligence tools in financial analysis. It also strengthened my understanding of how data visualisation can support preliminary transaction monitoring and risk analysis while recognising that further investigation is necessary to confirm suspicious activity.